WORKED SOLUTION

STEP II 1995, Question 5

Pure • Optimisation and Concavity

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Read the question before you go on. It is in the STEP Support Programme's question database:

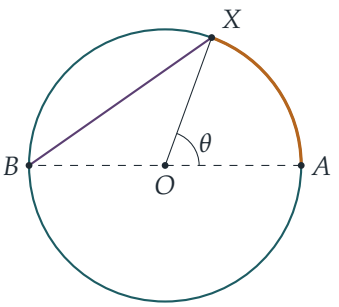
step.maths.org/questions/95-s2-q5

The key idea

Choices (i) and (iii) are the two ends of the family in (ii): $\theta \to 0$ means swimming straight across and $\theta \to \pi$ means running all the way. So we write the total time as a function of θ on $[0, \pi]$ and ask where its minimum is. The surprise is that the function is concave, so its only stationary point is a maximum, and the best route is always at one end.

Solution

Take the radius of the pool as 1 unit of length and her swimming speed as 1, so that she runs at speed k. Times are then in units of (radius)/(swimming speed), which doesn't affect any comparison.



The three times

(i) She swims the diameter AB, of length 2, taking time 2.

(iii) She runs a semicircle, of length π, taking time π/k.

(ii) She runs the arc AX, of length θ (radius 1), in time θ/k. Since $\angle AOB = \pi$, we have

$\angle XOB = \pi - \theta$. Triangle XOB is isosceles with $OX = OB = 1$, so the chord is

$$XB = 2\sin\tfrac{\pi-\theta}{2} = 2\cos\tfrac{\theta}{2}.$$

(Equivalently, $\angle AXB = \frac{\pi}{2}$ because AB is a diameter, and $\angle XAB = \frac{\pi-\theta}{2}$ from the isosceles triangle AOX, so $XB = 2\sin\angle XAB$.) Her total time is

$$T(\theta) = \frac{\theta}{k} + 2\cos\frac{\theta}{2}, \qquad 0 < \theta < \pi.$$

This formula also makes sense at the ends: $T(0) = 2$ is the time for (i) and $T(\pi) = \pi/k$ is the time for (iii). So all three choices are described by one continuous function on $[0, \pi]$.

Where is T smallest?

Differentiating,

$$T'(\theta) = \frac{1}{k} - \sin\frac{\theta}{2}, \qquad T''(\theta) = -\tfrac{1}{2}\cos\frac{\theta}{2}.$$

For $0 < \theta < \pi$ we have $0 < \frac{\theta}{2} < \frac{\pi}{2}$, so $\cos\frac{\theta}{2} > 0$ and $T''(\theta) < 0$: the function T is strictly concave on $[0, \pi]$.

> **Ask yourself** If $k > 1$ there is a stationary point at $\sin\frac{\theta}{2} = \frac{1}{k}$. Isn't that the best place to dive in?

No: since $T'' < 0$ there, it is a *maximum*, the worst place to dive in. More generally, a strictly concave function on a closed interval lies strictly above the chord joining its end values, so for every θ with $0 < \theta < \pi$,

$$T(\theta) > \Big(1 - \frac{\theta}{\pi}\Big)T(0) + \frac{\theta}{\pi}\,T(\pi) \geq \min(T(0),\, T(\pi)).$$

So whatever the value of k, some interior choice of X is never even as quick as the better of (i) and (iii). (If $k \leq 1$ we can see this even more directly: then $T'(\theta) > 0$ on $(0, \pi)$, as $\sin\frac{\theta}{2} < 1 \leq \frac{1}{k}$, so T is increasing and swimming straight away is best.)

Comparing (i) and (iii)

It remains to compare $T(0) = 2$ with $T(\pi) = \pi/k$:

- if $k < \frac{\pi}{2}$, then $\pi/k > 2$, so she should **swim directly**, choice (i);
- if $k > \frac{\pi}{2}$, then $\pi/k < 2$, so she should **run all the way round**, choice (iii);
- if $k = \frac{\pi}{2}$, (i) and (iii) take the same time, and either is best.

The physical reason is simply that the semicircle is $\frac{\pi}{2}$ times as long as the diameter.

Finally, choice (ii) is the one she will never take: for every $k > 0$ and every $\theta \in (0, \pi)$ it is strictly slower than the better of (i) and (iii).

Checks

With $k = 3$, running takes $\pi/3 \approx 1.047$, beating swimming's 2; the stationary point is at $\theta = 2\arcsin\frac{1}{3} \approx 0.680$, where $T \approx 2.112$, which is slower than either end, as a maximum should be. A computer evaluating T on a fine grid of θ for thousands of random k between e^{-4} and e^{4} never found an interior value below $\min(2, \pi/k)$, and confirmed that the changeover is at $k = \pi/2 \approx 1.571$.

Common slips

- Setting $T'(\theta) = 0$ and declaring the result the optimum without checking its nature: here it is a maximum.
- Using the arc XB instead of the chord, or getting the chord as $2\sin\frac{\theta}{2}$ (that is AX, not XB).
- Forgetting the case $k \leq 1$, where there is no stationary point at all.

Beyond STEP

The principle at work is that a concave function on an interval attains its minimum at an endpoint; in higher dimensions this becomes the fact that a linear (or concave) objective over a polygon is minimised at a vertex, which is the foundation of linear programming. The question is also a cousin of Snell's law and Fermat's principle of least time, where the answer usually is an interior point because the time function is convex instead. For more problems of this flavour, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.